

DBMS — Concurrency Problems

Comprehensive Lecture Notes & Exam Revision Guide

Concurrency refers to multiple transactions executing simultaneously with interleaved operations. When transactions run in parallel without adequate concurrency control, inter-transaction interference leads to data anomalies.

MASTER MEMORY TRICK

D I L U P

Dirty Read • Incorrect Summary • Lost Update • Unrepeatable Read • Phantom Read

1. DIRTY READ PROBLEM

(UNCOMMITTED READ / READ-AFTER-WRITE)

A **Dirty Read** occurs when a transaction reads a data value that has been modified by another concurrent transaction but is *not yet committed*. If the writing transaction later rolls back, the read data becomes completely invalid.

Execution Sequence

DIRTY READ

```
T1  READ(A) → 100
T1  WRITE(A) → 50 [Uncommitted]
T2  READ(A) → 50 [Reads Dirty Value!]
T2  COMPUTE: 50 - 10 = 40; COMMIT
T1  FAILURE! → ROLLBACK ABORT
```

Result: T1 reverts A to 100, but T2 already processed invalid value 50!

Key takeaways

- **Definition:** Reading uncommitted data written by another transaction.
- **Alias:** Uncommitted Read / WR Conflicts.
- **Core Cause:** Lack of shared/exclusive read-write locks.

 **Memory Trick:** Dirty = Uncommitted Read!

2. INCORRECT SUMMARY PROBLEM

(INCONSISTENT ANALYSIS)

Occurs when one transaction calculates an aggregate summary (e.g., SUM, AVG, COUNT) over a set of records while another transaction is actively modifying those records, producing an inconsistent total.

Aggregate Timeline

INCONSISTENT

```
Init  A = 1000, B = 1000
T1    A = A - 50 → 950
T2    READ(A) → 950 [Intermediate]
T2    CALCULATES AVG = 983 [Incorrect!]
T1    B = B + 50 → 1050; COMMIT
```

Key Takeaways

- **Definition:** Aggregate query reads intermediate states during concurrent updates.
- **Affected Ops:** SUM, AVG, COUNT, MIN/MAX.
- **Result:** Violates overall database balance integrity.

 **Memory Trick:** Summary = Aggregate on changing data!

3. LOST UPDATE PROBLEM

(OVERWRITTEN UPDATES)

Occurs when two transactions read the same initial data item and both update it. The transaction that writes last overwrites the update made by the earlier transaction, causing that update to be lost.

Overwriting Sequence

DATA LOST

```
Init  Likes = 5
T1    READ Likes → 5
T2    READ Likes → 5
```

Key Takeaways

- **Definition:** Successive writes overwrite earlier committed modifications.
- **Real-World Impact:** E-commerce inventory errors, duplicate item counters, lost balance deposits.

4. UNREPEATABLE READ PROBLEM

(VALUE CHANGED BETWEEN READS)

Occurs when a transaction reads the **same row/value twice** during its execution, but receives different values because another transaction modified and committed that row between the two reads.

Value Change Sequence

DIFFERENT VALUES

T1 READ(X) → 100 [First Read]
T2 UPDATE X = 50; COMMIT
T1 READ(X) → 50 [Second Read!]

Result: T1 reads X twice and gets two completely different values (100 ≠ 50).

Key Characteristics

- **Target:** Applies to the *same existing data row/value*.
- **Condition:** Intervening transaction **COMMITTS** its changes (unlike Dirty Read).

 **Memory Trick:** Unrepeatable = Same READ row, different value!

5. PHANTOM READ PROBLEM

(ROW APPEARS/DISAPPEARS)

Occurs when a transaction re-executes a range query (e.g., WHERE salary > 50000) and finds that the **set of rows satisfying the condition has changed** because another transaction inserted or deleted rows.

Phantom Insertion Sequence

NEW PHANTOM ROW

T1 SELECT WHERE salary > 50k → 3 Rows
T2 INSERT Emp4 (salary=60k); COMMIT
T1 SELECT WHERE salary > 50k → 4 Rows!

Result: A new row "phantom" appears in T1's result set during re-querying.

Key Characteristics

- **Target:** Applies to *result sets / collections of rows*.
- **Cause:** Concurrent INSERT or DELETE operations in predicate range.

 **Memory Trick:** Phantom = Row appears/disappears in query result!

ALL CONCURRENCY PROBLEMS AT A GLANCE

Problem	Core Description	Memory Key
Dirty Read	Reads uncommitted modifications of another transaction	Dirty = Uncommitted
Incorrect Summary	Aggregate calculation runs while data is actively changing	Summary = Aggregate
Lost Update	A committed update is overwritten by a concurrent write	Update Lost
Unrepeatable Read	Re-reading the <i>same row</i> yields a modified value	Same Row ≠ Same Value
Phantom Read	Re-running a <i>range query</i> yields new/missing rows	Phantom = Row Count Changed

CRITICAL DIFFERENCES FOR EXAMS

Dirty Read vs. Unrepeatable Read

Dirty Read	Unrepeatable Read
Reads uncommitted data	Reads committed data
Other transaction may rollback	Other transaction commits update

Unrepeatable Read vs. Phantom Read

Unrepeatable Read	Phantom Read
Single existing row value changes	Result set row count changes
Focus on UPDATE ops	Focus on INSERT / DELETE ops